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Primary Knee

Logistic Regression and Machine Learning Models Cannot Discriminate Between Satisfied and Dissatisfied Total Knee Arthroplasty Patients



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ABSTRACT

Background: Approximately 20% of total knee arthroplasty (TKA) patients are found to be dissatisfied or unsure of their satisfaction at 1-year post-surgery. This study attempted to predict 1-year post-surgery dissatisfied/unsure TKA patients with pre-surgery and surgical variables using logistic regression and machine learning methods.

Methods: A retrospective analysis of patients who underwent primary TKA for osteoarthritis between 2012 and 2016 at a single institution was completed. Patients were split into satisfied and dissatisfied/unsure groups. Potential predictor variables included the following: demographic information, patella resurfaced, posterior collateral ligament sacrificed, and subscales from the Knee Society Knee Scoring System, the Knee Society Clinical Rating System, the Western Ontario and McMaster Universities Osteoarthritis Index, and the 12-Item Short Form Health Survey version 2. Logistic regression and 6 different machine learning methods were used to create prediction models. Model performance was evaluated using discrimination (AUC [area under the receiver operating characteristic curve]) and calibration (Brier score, Cox intercept, and Cox slope) metrics.

Results: There were 1432 eligible patients included in the analysis, 313 were considered to be dissatisfied/unsure. When evaluating discrimination, the logistic regression (AUC = 0.736) and extreme gradient boosted tree (AUC = 0.713) models performed best. When evaluating calibration, the logistic regression (Brier score = 0.141, Cox intercept = 0.241, and Cox slope = 1.31) and gradient boosted tree (Brier score = 0.149, Cox intercept = 0.054, and Cox slope = 1.158) models performed best.

Conclusion: The models developed in this study do not perform well enough as discriminatory tools to be used in a clinical setting. Further work needs to be done to improve the performance of pre-surgery TKA dissatisfaction prediction models.

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Approximately 20% of patients are dissatisfied or unsure of their satisfaction with their total knee arthroplasty (TKA) [1,2]. This is a very popular surgery with approximately 75,000 TKAs performed

in 2018–2019 in Canada [3]. This means that in 2018–2019 alone there were approximately 15,000 TKA patients dissatisfied or unsure. This is a substantial problem. If dissatisfied and unsure patients can reliably be identified pre-surgery, specific interventions or the optimization of non-operative modalities could result in a reduction in dissatisfied or unsure patients.

There have been a number of previous attempts at predicting 1-year and 2-year TKA dissatisfaction using pre-surgery and surgical variables [4–7]. Several studies have used logistic regression [7] and ordinal logistic regression [6]. These attempts have either not been successful or failed to generalize to other samples [6,7].

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Recently, machine learning prediction models have been used to determine if they can outperform logistic regression, with Kunze et al (2020) [8] and Farooq et al (2020) [4] both finding that machine learning methods were superior to logistic regression. Although there has been a recent proliferation in the use of machine learning methods to predict outcomes in arthroplasty and orthopedic medicine in general [9,10], these studies represent the first attempts at employing machine learning models to predict TKA dissatisfaction. Despite the reported success of these studies, they have not exhaustively explored all potential predictor variables of TKA dissatisfaction. Specifically, neither of these studies included patient-reported outcome measures (PROMs)

When evaluating prediction models, it is also important to consider their intended use. If the models are meant to be used as standalone discriminatory tools, then it must be established what kind of performance is necessary for the model to be considered clinically useful. However, if models are intended to be used to provide supplemental information and to help inform surgeon and patient decision-making, then it may be more important to evaluate a model's ability to generate an accurate probability that a patient will be dissatisfied than its discriminatory capacity.

The purpose of this study is to evaluate the ability of a standard logistic regression model and models created using machine learning methods to predict TKA dissatisfaction 1-year post-surgery using pre-surgery and surgical variables that can be considered as typically collected as part of a pre-surgery assessment. This study will also aim to evaluate if the variables included in this analysis contain the necessary information to predict dissatisfaction or if additional variables or analysis techniques may be required. Both types of models were evaluated by their ability to discriminate between satisfied and dissatisfied or unsure patients and their ability to generate accurate probabilities that patients will be dissatisfied or unsure. Because of the findings of Kunze et al (2020) and Farooq et al (2020), it was expected that the machine learning models would outperform standard logistic regression.

Methods

Participants

This study is a retrospective analysis of prospectively collected data that included 1432 patients who underwent a primary TKA at a single institution between August 2012 and December 2016. To be included in the study patients were required to have been diagnosed with osteoarthritis, have no missing demographics information, and have completed both a pre-surgery and 1-year post-surgery Knee Society Knee Scoring System (KSS) questionnaire [11]. This study received ethics approval from the institutional ethics review board.

Outcome Measures

The primary outcome measure was a dichotomized version of the KSS satisfaction subscale at 1-year post-surgery. This subscale includes 5 questions that evaluate a patient's satisfaction with their pain while sitting, pain while lying down, ability to get out of bed, ability to perform light household duties, and ability to perform recreational activities. Possible response options for each question (score) are very satisfied (8), satisfied (6), neutral (4), dissatisfied (2), and very dissatisfied (0). These 5 questions are summed together to create a score from 0 to 40 (worst to best). Satisfaction scores were dichotomized into satisfied and dissatisfied groups. Patients with a satisfaction score ≤ 24 were classified as dissatisfied or neutral and patients with a satisfaction score >24 were considered as satisfied. A cut-point of 24 was chosen as this would mean

that a patient must have responded to a minimum of one of the satisfaction questions as dissatisfied or 3 questions as neutral.

Predictor Variables

Candidate predictor variables included the following: age, gender, body mass index (BMI), American Society of Anesthesiologists (ASA) class, alcohol consumption, smoking status, living status, previous knee surgeries, pre-surgery knee alignment, patella re-surfaced, posterior collateral ligament sacrificed, as well as subscales from the following questionnaires: the KSS, Pre-op and Post-op versions [11], the Knee Society Clinical Rating System (KSCRS) [12], the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) [13], and the 12-Item Short Form Health Survey version 2 (SF-12) [14]. ASA class, alcohol consumption, smoking status, living status, and previous knee surgeries were dichotomized. ASA class was split into 2 categories, patients scored as class 1 and 2, and patients scored as class 3 and 4. Alcohol consumption was split into medium and light (<5 drinks per week) and heavy drinkers (>5 drinks per week). Living status was split into patients who lived alone and patients who lived with others. Previous surgery was split into patients who had previously undergone knee surgery and those who had not.

Missing Data

Any variables that had greater than 15% of the data missing were not included in the analysis. The KSCRS anteroposterior stability, mediolateral stability, flexion contracture, extension lag, and valgus alignment were not included in the analysis, as too much data were missing. Missing data were imputed using k nearest neighbors' imputation [15].

Statistical Analysis

The data were split 70:30 between a training ($n = 1022$) and a test set ($n = 437$). The split was stratified on the dissatisfaction variable to ensure that there was roughly the same ratio of dissatisfied patients in the training and the test sets. Seven different statistical and machine learning models were attempted. These included: standard logistic regression, elastic net logistic regression [16], random forest [17], gradient boosted trees [18], extreme gradient (XG) boosted trees [19], support vector machines (SVM) [20], and neural networks [21]. For all models except neural networks 10-fold cross validation was used to select the optimal model. For neural networks, a single validation set was used.

The standard logistic regression model was selected using backwards elimination. A P -value of .20 was used as the cutoff so that no potentially important variables were excluded. Variance inflation factor (VIF) was calculated for the model, with a variance inflation factor >5 being considered problematic.

Elastic net logistic regression adds regularization to a standard logistic regression model to avoid overfitting and improve model performance [22,23].

A random forest is an ensemble tree-based method [22,23]. This method fits a large number of classification trees (usually several thousand) using bootstrapped samples and a random subset of predictor variables, preventing classification trees from being too uniform. The outcome of each of the individual trees is aggregated and a final prediction is generated [22,23].

Gradient boosting and XG boosting are both gradient boosting methods; however, they use different algorithms to select and optimize models [19]. Gradient and XG boosted trees, like random forests, are ensemble tree-based methods. However, these methods fit a series of classification trees sequentially, with each tree created

attempting to fix the prediction errors made by the previous tree. This process is typically repeated several hundred or several thousand times.

SVMs are models that attempt to draw a hyperplane that maximizes the distance between the 2 classes [22,23]. This model type can be linear or non-linear. To create a non-linear model, a kernel can be used, to allow for greater flexibility. This study used a non-linear version of SVMs. Platt scaling was used to calibrate the SVM model [22,23].

Neural network models are meant to emulate the neuronal circuitry of the human brain [23]. Neural networks consist of a series of inputs, hidden layers of nodes, and an output. Data must pass through all the layers and a prediction is generated at the end. These models are extremely flexible and can map complex patterns [23].

Model Evaluation

After the optimal model for each type of machine learning algorithm was generated using the training set, its performance was evaluated using the test set. Metrics that evaluated model discrimination and calibration were used.

Discrimination

Discrimination is the model's ability to accurately predict a patient's satisfaction status (ie, correctly predicting that a dissatisfied patient would be dissatisfied). Model discrimination was evaluated using area under the receiver operating characteristic

curve (AUC). Additionally, cut-points were established that maximized model accuracy and overall accuracy, sensitivity, specificity, positive predictive value, negative predictive value, and kappa scores were calculated [23].

Calibration

Calibration is the model's ability to accurately assign probabilities to a patient's outcome (ie, patients who are assigned a probability of 0.6 are dissatisfied 60% of the time). The models' calibration was evaluated using the Brier score, Cox intercept, and Cox slope. Brier score is a measure of how closely probabilities predicted by the model match the actual outcome [23], with a lower score indicating better performance. Cox intercept and slope are measures of how well the calibration probability matches with an actual outcome. The optimal Cox intercept is 0 and the optimal Cox slope is 1, with values close to 0 and 1 indicating good overall calibration. Calibration plots were generated for the models that performed best [23].

Model Interpretation

Not all machine learning models are equally interpretable or provide information about variable importance. If the algorithm allowed for straightforward interpretation of the importance or effect of individual variables on model performance, then that information was included for the best performing models. For logistic regression an output table including odds ratios was generated. For tree-based models, relative influence plots were created, and

Table 1
Descriptive Statistics Including Mean, Standard Deviation, and *t*-Test/Chi-Squared *P*-Values.

	Dissatisfied or Neutral (n = 313)		Satisfied (n = 1119)		<i>t</i> -Test/Chi-Squared
	Mean/Frequency	SD	Mean/Frequency	SD	
Age	66.62	10.88	66.79	9.418	.809
BMI	33.77	8.241	33.347	7.244	.106
Gender (female)	67.7%		63.5%		.12
Living status (live alone)	25.6%		21.3%		.023
Alcohol consumption (>5 drinks/wk)	15.7%		21.1%		.206
Smoker	47.6%		51.7%		.884
ASA class (3 and 4)	60.7%		54.8%		.171
PCL sacrificed	85.3%		87.3%		.353
Patella resurfaced	22.0%		27.3%		.059
Valgus alignment (>10° valgus)	28.4%		24.2%		.128
Varus alignment (<2° valgus)	61.0%		72.1%		<.001
Neutral alignment (2°–10° valgus)	10.5%		3.7%		<.001
KSCRS Pre-op Pain (0–50)	19.948	6.666	21.527	6.578	<.001
KSCRS Pre-op Range of Motion Extension	5.054	9.356	4.415	5.767	.033
KSCRS Pre-op Range of Motion Flexion	102.326	13.846	103.849	13.737	.205
KSCRS Pre-op Total Function (–20 to 100)	45.589	12.363	48.453	12.36	.883
SF-12 Pre-op MCS (0–100)	49.772	10.978	54.01	10.18	.062
SF-12 Pre-op PCS (0–100)	30.265	7.55	32.187	8.33	<.001
WOMAC Pre-op Pain (0–100)	42.824	16.044	49.805	17.438	<.001
WOMAC Pre-op Stiffness (0–100)	38.068	19.804	42.095	19.941	.002
WOMAC Pre-op Function (0–100)	42.693	15.436	49.932	16.724	<.001
WOMAC Pre-op Total Score (0–100)	41.741	14.702	48.191	15.576	<.001
KSS Symptoms (0–25)	7.019	4.814	8.712	5.345	<.001
KSS Pre-op Satisfaction (0–40)	12.231	6.595	14.725	7.275	<.001
KSS Pre-op Expectations (3–15)	13.198	2.216	13.727	1.667	<.001
KSS Walking Standing (–10 to 30)	10.498	7.655	12.061	7.577	.001
KSS Standard Activity (0–30)	10.997	4.842	12.633	4.98	<.001
KSS Advanced Activity (0–25)	4.402	4.03	5.368	4.299	<.001
KSS Discretionary Activity (0–15)	4.824	3.141	5.561	3.159	<.001

Values in bold indicate a *P*-value of less than .05.

BMI, body mass index; ASA, American Society of Anesthesiologists; PCL, posterior collateral ligament; KSCRS, Knee Society Clinical Rating System; SF-12, 12-Item Short Form Health Survey version 2; MCS, mental component score; PCS, physical component score; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; KSS, Knee Society Knee Scoring System; SD, standard deviation.

Table 2
Discrimination and Calibration Measures of the Test Set Including AUC, Accuracy, Sensitivity, Specificity, PPV, NPV, Kappa Score, Brier Score, Cox Intercept, and Cox Slope.

	Standard Logistic Regression	Random Forest	Gradient Boosted Trees	XG Boosted Trees	Support Vector Machines	Neural Networks
Discrimination						
AUC	0.736	0.709	0.689	0.713	0.6	0.702
Accuracy	0.795	0.786	0.79	0.781	0.664	0.804
Sensitivity	0.155	0.202	0.083	0.214	0.44	0.131
Specificity	0.951	0.928	0.962	0.919	0.719	0.968
Positive predictive value	0.433	0.405	0.35	0.391	0.276	0.5
Negative predictive value	0.822	0.827	0.812	0.828	0.841	0.821
Kappa score	0.139	0.16	0.064	0.161	0.13	0.137
Calibration						
Brier score	0.141	0.147	0.149	0.165	0.243	0.153
Cox intercept	0.241	0.473	0.054	0.798	−1.246	0.112
Cox slope	1.310	1.092	1.158	3.14	0.098	1.663

Cut-points used to generate accuracy, sensitivity, specificity, PPV, NPV, and kappa score were chosen to maximize accuracy.

XG, extreme gradient; AUC, area under the receiver operating characteristic curve; PPV, positive predictive value; NPV, negative predictive value.

variables were grouped into high, medium, and low influence categories.

All analyses were completed in R version 4.0.3 [24].

Results

Overall, 1432 patients met the inclusion criteria. The study population had a mean age of 66.75 and 923 (64.5%) were female. A total of 313 (21.8%) patients were classified as dissatisfied or unsure at 1-year post-surgery. Descriptive statistics including *t*-tests/chi-squares are included in Table 1.

Model Performance

Discrimination

Measures of discrimination are included in Table 2. The models that performed best when considering discrimination metrics were the standard logistic regression and XG boosted tree models. The logistic regression model outperformed the elastic net logistic regression model and was chosen as the final model. When considering the AUC, accuracy, sensitivity, specificity, and kappa score, the logistic regression model performed best.

Calibration

Measures of model calibration are included in Table 2. When considering Brier score, Cox intercept, and Cox slope, the best performing models were the logistic regression and gradient boosted tree models (Fig. 1A and 1C). The XG boosted tree model was very poorly calibrated (Fig. 1B). This is illustrated by the proximity of the dotted line and the triangles to the thick gray line.

Model Interpretation

The output of the logistic regression model is included in Table 3, outlining important predictors. The relative influence of the individual variables for the XG boosted and gradient boosted models has been included in Figure 2, with variables being grouped into high, medium, and low influence categories.

Discussion

When considering AUC, the best and second-best performing models respectively were standard logistic regression and XG boosted trees. This finding was unexpected and suggests that when predicting dissatisfaction using variables that can be considered as typically collected as part of a pre-surgery assessment, more complex machine learning models may not offer a predictive advantage over standard logistic regression. This was contrary to

the findings of Kunze et al (2020) and Farooq et al (2020) who found that machine learning models outperformed standard logistic regression when predicting dissatisfaction. However, this finding was consistent with a systematic review by Christodoulou et al (2019), which found in a medical context machine learning methods did not outperform logistic regression [25]. A possible reason for the disparity is that this study made use of different variables than Kunze et al (2020) and Farooq et al (2020). Notably, we included the SF-12 and WOMAC as predictor variables. These questionnaires are validated and commonly used with arthroplasty patients. Additionally, a different satisfaction measure was used as the dependent variable. Although the variables used in this analysis have historically been considered as important, it appears that they do not contain the necessary information to accurately predict 1-year post-surgery TKA dissatisfaction and are not capable of fully realizing the benefits of machine learning methods. A wider variety of explanatory variables and analytical methods must be investigated.

Despite acceptable AUC values, when establishing discriminatory cut-points and examining accuracy, sensitivity, specificity, and kappa scores none of the models performed particularly well. How a discriminatory cut-point is chosen depends on the priority of the decision maker. However, in this study cut-points were simply chosen to maximize accuracy. Generally, models performed at or below the no information rate. This means that when maximizing accuracy, the models' performance was approximately the same or worse as predicting that all patients would be satisfied. Additionally, the models have low sensitivity scores, indicating that they struggle to accurately identify dissatisfied patients. It is possible to improve sensitivity of a model by adjusting the cut-point that is chosen. However, this would come at the expense of specificity and overall accuracy.

When evaluating these models, it is also important to consider how the model is going to be used and how well the model needs to perform in order to be clinically useful as a discriminatory tool. This would require a discussion of acceptable sensitivity and specificity values. It would also require that a judgment be made about which is more costly, not performing surgery on a patient who would be satisfied or performing surgery on a patient who will be dissatisfied. The machine learning models presented in this paper likely do not discriminate well enough to be useful as clinical decision tools.

Despite the poor discriminatory capacity of the models, the calibration scores appear to be more promising. This indicates that the models may be better at generating an accurate probability that a patient will be dissatisfied than discriminating between satisfied and dissatisfied patients. Although not optimal, the best calibrated models were the logistic regression and gradient boosted tree models. It would be possible to use the probabilities generated by

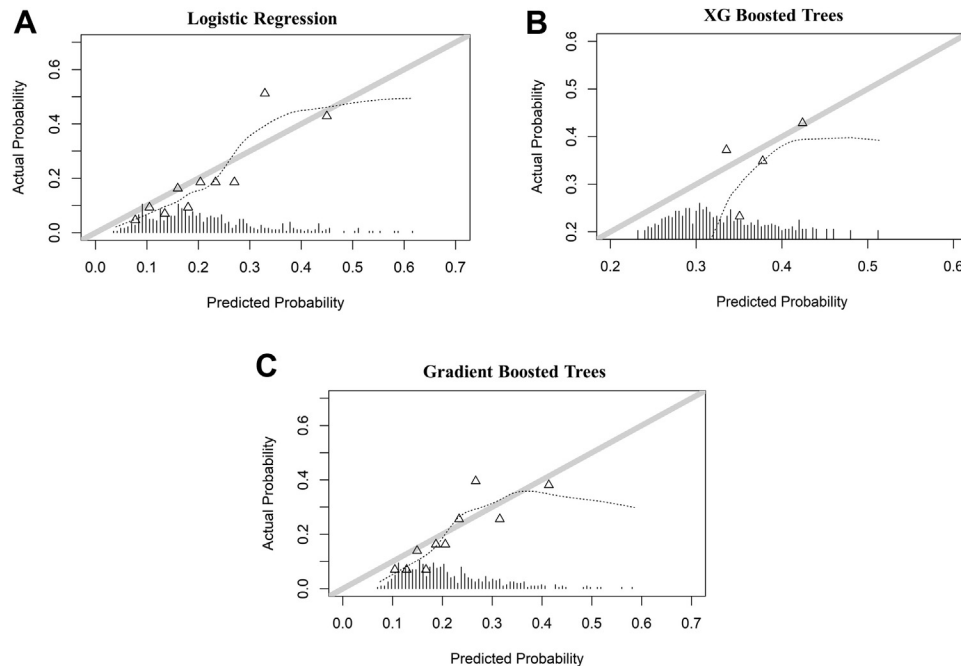


Fig. 1. Calibration plots for the logistic regression (A), XG boosted trees (B), and gradient boosted tree models. The triangles indicate a group of 40 patients. The average probability for the group is taken and then that is compared to percentage of patients who are actually dissatisfied. The dotted line is a smoothed calibration line. The thick gray line indicates perfect calibration. XG, extreme gradient.

these models to help inform surgeon and patient decision-making, without the models making an explicit prediction about whether a patient is dissatisfied or not. Harrell (2015) has argued that there can be great value in using medical prediction models in this probabilistic fashion. This approach has the additional benefit of allowing the surgeon's judgment and the patient's viewpoint to be used in treatment decisions [26,27]. In this way information that has not been incorporated into a prediction model can be included in the decision-making process.

When evaluating significant and influential predictors of satisfaction, a consistent theme appeared in the logistic regression, gradient boosted trees, and XG boosted trees models. Specifically, age, pre-surgery WOMAC function, pre-surgery SF-12 MCS, and

pre-surgery patient expectations were all identified as significant/influential predictors of satisfaction. It is important to note that unlike logistic regression, the relative influence of a variable does not provide any information about the direction or the context in which the variable is influencing the prediction. For example, in the logistic regression model the odds ratio for age indicates that the older a patient is the more likely they are to be dissatisfied. This relationship cannot be assumed when examining the relative influence plot. The information from logistic regression and tree-based models in some sense can be considered complimentary as they are providing different types of information about predictors of dissatisfaction and if used together could potentially help further our understanding of how these variables influence satisfaction. In

Table 3
Logistic Regression Model Output Table.

Predictors	Dissatisfaction				
	Estimates	Standard Error	P-Value	Odds Ratios	CI
(Intercept)	3.462	1.153			
Age	0.017	0.009	.049	1.02	1.00-1.03
BMI	0.002	0.011	.818	1.00	0.98-1.02
Gender (male)	−0.021	0.174	.901	0.98	0.69-1.37
Valgus alignment	−0.609	0.332	.066	0.54	0.29-1.05
Varus alignment	−0.856	0.311	.006	0.42	0.23-0.79
Patella resurfaced	−0.37	0.185	.045	0.69	0.48-0.99
KSCRS Pain	−0.018	0.013	.147	0.98	0.96-1.01
SF-12 MCS	−0.0277	0.008	.001	0.97	0.96-0.99
SF-12 PCS	−0.018	0.012	.149	0.98	0.96-1.01
WOMAC Function	−0.014	0.006	.032	0.99	0.97-1.00
KSS Pre-surgery Satisfaction	−0.016	0.015	.294	0.98	0.96-1.01
KSS Pre-surgery Expectations	−0.155	0.04	<.001	0.86	0.79-0.93
KSS Walking Standing	0.022	0.012	.072	1.02	1.00-1.05
Observations	1003				

Age, BMI, and gender were forced into the model regardless of significance.

BMI, body mass index; CI, confidence interval; KSCRS, Knee Society Clinical Rating System; SF-12, 12-Item Short Form Health Survey version 2; MCS, mental component score; PCS, physical component score; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; KSS, Knee Society Knee Scoring System.

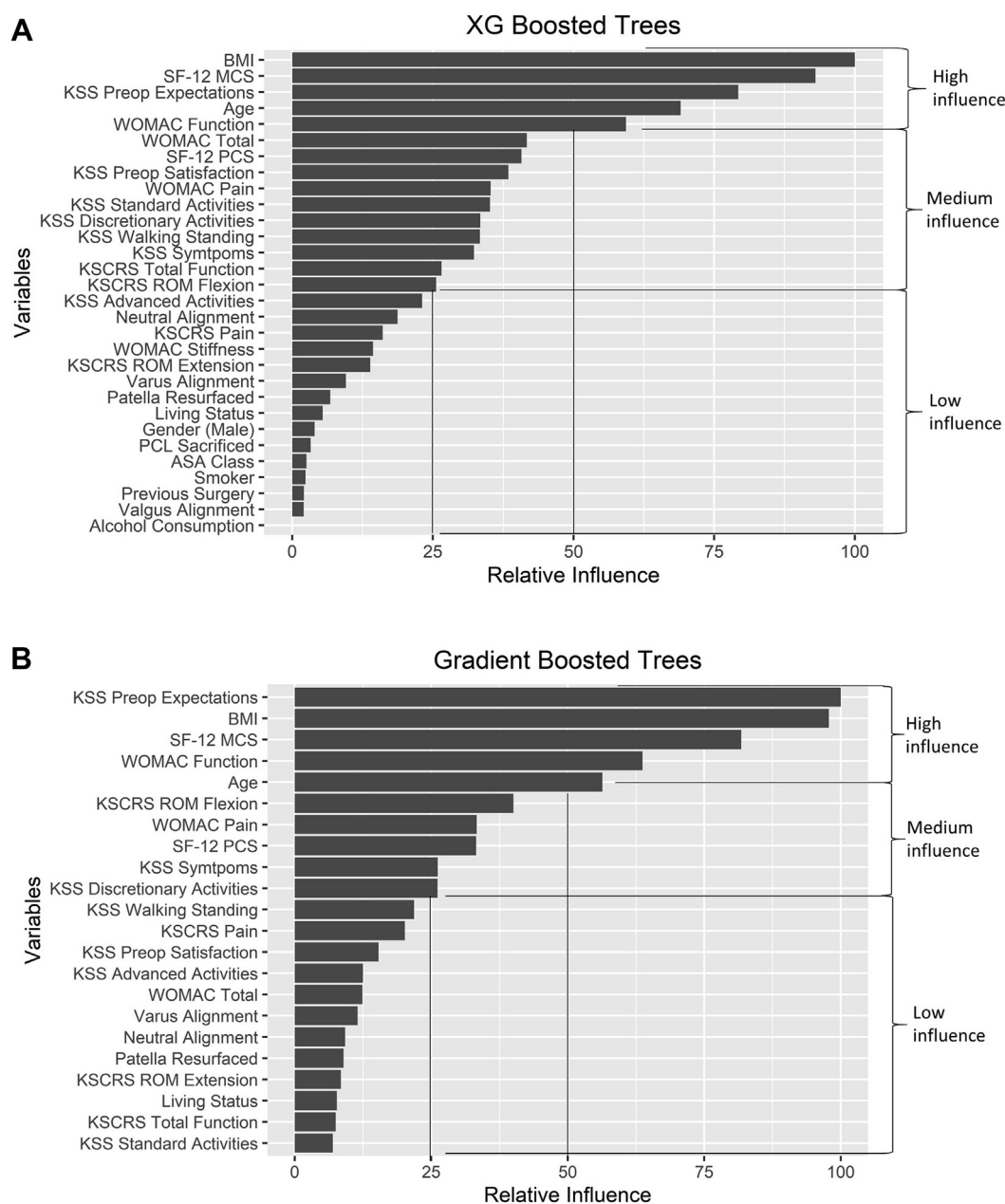


Fig. 2. This is a graphical representation of relative importance of variables for the extreme gradient boosted trees (A) and gradient boosted trees (B) scaled from 0 to 100, with the most important variable being scored as 100 and the least important scored as 0. BMI, body mass index; SF-12, 12-Item Short Form Health Survey version 2; MCS, mental component score; KSS, Knee Society Knee Scoring System; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; PCS, physical component score; KSCRS, Knee Society Clinical Rating System; PCL, posterior collateral ligament; ASA, American Society of Anesthesiologists; ROM, range of motion.

the gradient boosted trees and XG boosted trees models, BMI was a highly influential variable, but was not a significant predictor in the logistic regression model. This finding does not necessarily indicate that higher BMI is associated with dissatisfaction. It is possible that BMI has a complex relationship with dissatisfaction and is conditionally important. This could help explain the inconsistent relationship between BMI and satisfaction in the literature [28].

The variables included in this study do not appear to have the necessary information to accurately predict 1-year post-surgery dissatisfaction, and it may be possible that there is no reasonable way to predict dissatisfaction using pre-surgery and surgical variables. However, before making this determination there are other analytical methods and predictors to explore. Part of the value of machine learning methods is their ability to map complex patterns

and include a wider variety of data than standard statistical models. Natural language processing is a form of machine learning that can be used to process language and predict outcomes. Training natural language processing models to make use of the information in clinical notes to predict dissatisfaction is a possible avenue for future research. Using machine learning to analyze radiographic images to predict dissatisfaction is also an area to investigate. Finally, the collection of different PROMs that provide more granular information about the patient may allow for the models to be improved. Specifically, determining what a patient would define as a successful outcome for their surgery. This could include determining, not just pre-surgery expectations, but specific activities that are particularly important or that the patient is motivated to perform. Additionally, psychological factors (ie, personality, self-

efficacy, and locus of control), socioeconomic status, financial well-being, and social support remain mostly unexplored in the TKA literature. Including these or other variables that may interact with standard PROMs and can take advantage of the flexibility of machine learning methods could potentially improve the performance of prediction models.

This study had several limitations. First, the study is a retrospective analysis, although the PROMs were collected prospectively. It could be a potential limitation that this study only included a 1-year follow up, although a systematic review and meta-analysis completed by Ramkumar et al (2018) has found no difference in outcomes 1 and 2 years post-surgery for lower body total joint arthroplasty, so a 1-year follow-up time should be acceptable [29]. This study was missing a comorbidity measure, and additionally, several of the KSCRS subscores had to be removed because of missing data. The choice to dichotomize ASA class, alcohol consumption, smoking status, living status, and previous knee surgery variables may have adversely affected model performance. Finally, despite the relatively large sample size of this study, it is likely still too small to generate an optimal prediction model.

In conclusion, all the models in this study do not perform well as discriminatory tools and are early attempts at using machine learning models to predict dissatisfaction. A wider array of variables and analytical techniques need to be investigated to determine if machine learning models can more accurately predict TKA dissatisfaction. In order to make this determination a decision must also be made with regards to how these tools will be used. If the models will not be used in a stand-alone discriminatory capacity, there may be greater value in creating well-calibrated prediction models that generate accurate probabilities than models that are meant to be purely discriminatory.

References

- [1] Kahlenberg CA, Nwachukwu BU, Mclawhorn AS, Cross MB, Cornell CN, Padgett DE. Patient satisfaction after total knee replacement: a systematic review. *HSS J* 2018;14:192–201.
- [2] Gunaratne R, Pratt DN, Banda J, Fick DP, Hons M, Khan RJK, et al. Patient dissatisfaction following total knee arthroplasty: a systematic review of the literature. *J Arthroplasty* 2017;32:3854–60.
- [3] Canadian Institute for Health Information. Hip and Knee Replacements in Canada: CJRR Annual Statistics Summary. Ottawa ON: CIHI; 2018–2019.
- [4] Farooq H, Deckard ER, Ziemba-Davis M, Madsen A, Meneghini RM. Predictors of patient satisfaction following primary total knee arthroplasty: results from a Traditional statistical model and a machine learning algorithm. *J Arthroplasty* 2020;35:3123–30.
- [5] Kunze KN, Akram F, Fuller BC, Zabawa L, Sporer SM, Levine BR. Internal validation of a predictive model for satisfaction after primary total knee arthroplasty. *J Arthroplasty* 2019;34:663–70.
- [6] Baker PN, Rushton S, Jameson SS, Reed M, Gregg P, Deehan DJ. Patient satisfaction with total knee replacement cannot be predicted from pre-operative variables alone: a cohort study from the National Joint Registry for England and Wales. *Bone Joint J* 2013;95:1359–65.
- [7] Garriga C, Sanchez-Santos MT, Judge A, Perneger T, Hannouche D, Lübbecke A, et al. Development of a model predicting non-satisfaction 1 year after primary total knee replacement in the UK and transportation to Switzerland. *Sci Rep* 2018;8:1–8.
- [8] Kunze KN, Polce EM, Sadauskas AJ, Levine BR. Development of machine learning algorithms to predict patient dissatisfaction after primary total knee arthroplasty. *J Arthroplasty* 2020;35:3117–22. <https://doi.org/10.1016/j.arth.2020.05.061>.
- [9] Ramkumar PN, Haeberle HS, Bloomfield MR, Schaffer JL, Kamath AF, Patterson BM, et al. Artificial Intelligence and arthroplasty at a single institution: Real-World Applications of machine learning to Big data, value-based Care, Mobile health, and Remote patient Monitoring. *J Arthroplasty* 2019;34:2204–9.
- [10] Haeberle HS, Helm JM, Navarro SM, Karnuta JM, Schaffer JL, Callaghan JJ, et al. Artificial Intelligence and machine learning in lower Extremity arthroplasty: a review. *J Arthroplasty* 2019;34:2201–3.
- [11] Noble PC, Scuderi GR, Brekke AC, Sikorski A, Benjamin JB, Lonner JH, et al. Development of a new Knee Society scoring system. *Clin Orthop Relat Res* 2012;470:20–32.
- [12] Insall JN, Dorr D, Scott R, Scott W. Knee Society score Rationale. *Clin Orthop Relat Res* 1989;26:13–4.
- [13] Bellamy N, Buchanan W, Goldsmith C, Campbell J, Stitt L. Validation study of WOMAC: a health status instrument for measuring clinically important patient relevant outcomes to antirheumatic drug therapy in patients with osteoarthritis of the hip or knee. *J Rheumatol* 1988;15:1833–40.
- [14] Ware JE, Kosinski M, Keller SD. A 12-Item Short-Form Health Survey: construction of scales and preliminary tests of reliability and validity. *Med Care* 1996;34:220–33.
- [15] Torgo L. Data mining with R: learning with case studies. 1st ed. Boca Raton, FL: Taylor & Francis; 2010.
- [16] Friedman J, Hastie T, Tibshirani R. Regularization paths for generalized linear models via coordinate descent. *J Stat Softw* 2010;33:1–22.
- [17] Liaw A, Wiener M. Classification and regression by randomForest. *R News* 2002;2:18–22.
- [18] Greenwell B, Boehmke B, Cunningham J, GBM Developers. "Gbm: Generalized Boosted Regression Models." R Package Version. 2019. p. 2.
- [19] Chen T, He T, Benesty M, Khotilovich V. Package 'xgboost'. R version. 2019. p. 90.
- [20] Meyer D, Dimitriadou E, Hornik K, Weingessel A, Leisch F, Chang CC, Lin CC. E1071: misc functions of the department of statistics, probability theory group.; 2019. R package version. 2021. p. 1–6.
- [21] Allaire JJ, Chollet F. keras: R Interface to 'Keras'. R package version 2.2. 0.
- [22] James G, Witten D, Hastie T, Tibshirani R. An Introduction to Statistical Learning. New York: Springer Science+Business Media; 2013.
- [23] Kuhn M, Johnson K. Applied Predictive Modeling. New York: Springer Science+Business Media; 2013.
- [24] R Core Team. R: a language and environment for statistical computing. Vienna Austria: R Foundation for Statistical Computing; 2020.
- [25] Christodoulou E, Ma J, Collins GS, Steyerberg EW, Verbakel JY, Van Calster B. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *J Clin Epidemiol* 2019;110:12–22.
- [26] Harrell F. Regression Modeling Strategies. New York: Springer Science+Business Media; 2015.
- [27] Spiegelhalter DJ. Probabilistic prediction in patient management and clinical trials. *Stat Med* 1986;5:421–33.
- [28] Nakano N, Shoman H, Olavarria F, Matsumoto T, Kuroda R, Khanduja V. Why are patients dissatisfied following a total knee replacement? A systematic review. *Int Orthopaedics* 2020;44:1971–2007. <https://doi.org/10.1007/s00264-020-04607-9>.
- [29] Ramkumar PN, Navarro SM, Haeberle HS, Ng M, Piuze NS, Spindler KP. No difference in outcomes 12 and 24 Months after lower Extremity total joint arthroplasty: a systematic review and meta-analysis. *J Arthroplasty* 2018;33:2322–9.

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Erratum and Corrigendum

Corrigendum to “Logistic Regression and Machine Learning Models Cannot Discriminate Between Satisfied and Dissatisfied Total Knee Arthroplasty Patients [The Journal of Arthroplasty 37 (2022) 267-273]



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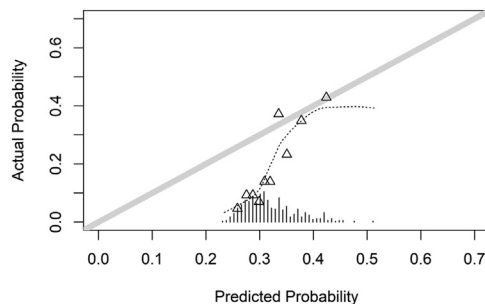


Fig. 1B. Calibration plot for the XG boosted trees model.

The authors would like to inform you that sections of both axes were missing from [Figure 1B](#) in their published article. The correct figure is provided below.

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